



Paper id: 250735

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Subject Code: RCA1804

Roll No:

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MCA-INT
(SEM VIII) THEORY EXAMINATION 2024-25
COMPUTER NETWORKS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A1. Attempt *all* questions in brief.

2 x 07 = 14

Q no.	Question
a.	Discuss the differences between simplex, half-duplex, and full-duplex communication.
b.	What is a router?
c.	Compare Guided and unguided transmission media.
d.	What do you mean by Multiplexing?
e.	Distinguish Between Unicast, Multicast, and Broadcast Transmission.
f.	Define the term ISDN.
g.	What are the Key Benefits of internet?

SECTION B2. Attempt any *three* of the following:

07 x 3 = 21

Q no.	Question
a.	Describe Computer Network. Explain various types of Computer Network.
b.	What are the different types of transmission media used in data communications? Explain with suitable example.
c.	What is switching? Describe various methods of Switching.
d.	Discuss about the techniques to improve QoS in internetworking.
e.	What are routing algorithms? Explain.

SECTION C3. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	With a neat block diagram explain the TCP/IP network model.
b.	Briefly explain Data Communication System and its characteristics.



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4. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	What do you mean by Signals impairment? Discuss various causes of impairment.
b.	Write short notes on a) Shannon capacity b) Nyquist bit rate

S. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	Explain Stop-and-Wait, Go-Back-N, and Selective-Repeat protocols.
b.	Briefly explain about Error detection techniques.

6. Attempt any *one* part of the following:

07 x 1 = 07

Q no.	Question
a.	What are protocols that make PPP powerful?
b.	Describe transition phase diagram of a PPP Connection.

7. Attempt any *one* part of the following:

$$07 \times 1 = 07$$

Q no.	Question
a.	Describe various connecting devices.
b.	What do you mean by Congestion control ? Explain various types of congestion control mechanisms? https://www.aktuonline.com